

Risk Metric Analysis Report

European Vanilla Option

Valuation Date: December 30, 2025

1. Product Parameters

Parameter	Value
Product Type	European Vanilla Option (CALL & PUT)
Strike Price	\$100.00
Time to Maturity	1.00 year
Option Types	CALL, PUT
Valuation Date	2025-12-30
Contract Multiplier	1

2. Market Data

Parameter	Value
Spot Price	\$100.00
Volatility	20.0%
Risk-Free Rate	5.0%
Dividend Yield	2.0%

3. Pricing Results

Metric	Call Option	Put Option
Premium	\$9.2270	\$6.3301
Intrinsic Value	\$2.8969	\$0.0000

Time Value	\$6.3301	\$6.3301
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Put-Call Parity Verification

The put-call parity relationship for European options with continuous dividends is: **$C - P = (S \cdot e^{(-qT)} - K \cdot e^{(-rT)}) \times M$** Where: C = Call premium = \$9.2270 P = Put premium = \$6.3301 S = Spot price = \$100.00 K = Strike price = \$100.00 q = Dividend yield = 2.0% r = Risk-free rate = 5.0% T = Time to maturity = 1.00 years M = Contract multiplier = 1 Verification: **LHS (C - P):\$2.8969 RHS ((S·e^(-qT)) - K·e^(-rT)) × M):\$2.8969 Difference:\$0.00000000**
Status: ✓ PUT-CALL PARITY HOLDS (Difference < 1e-6)

4. Risk Metrics (Greeks)

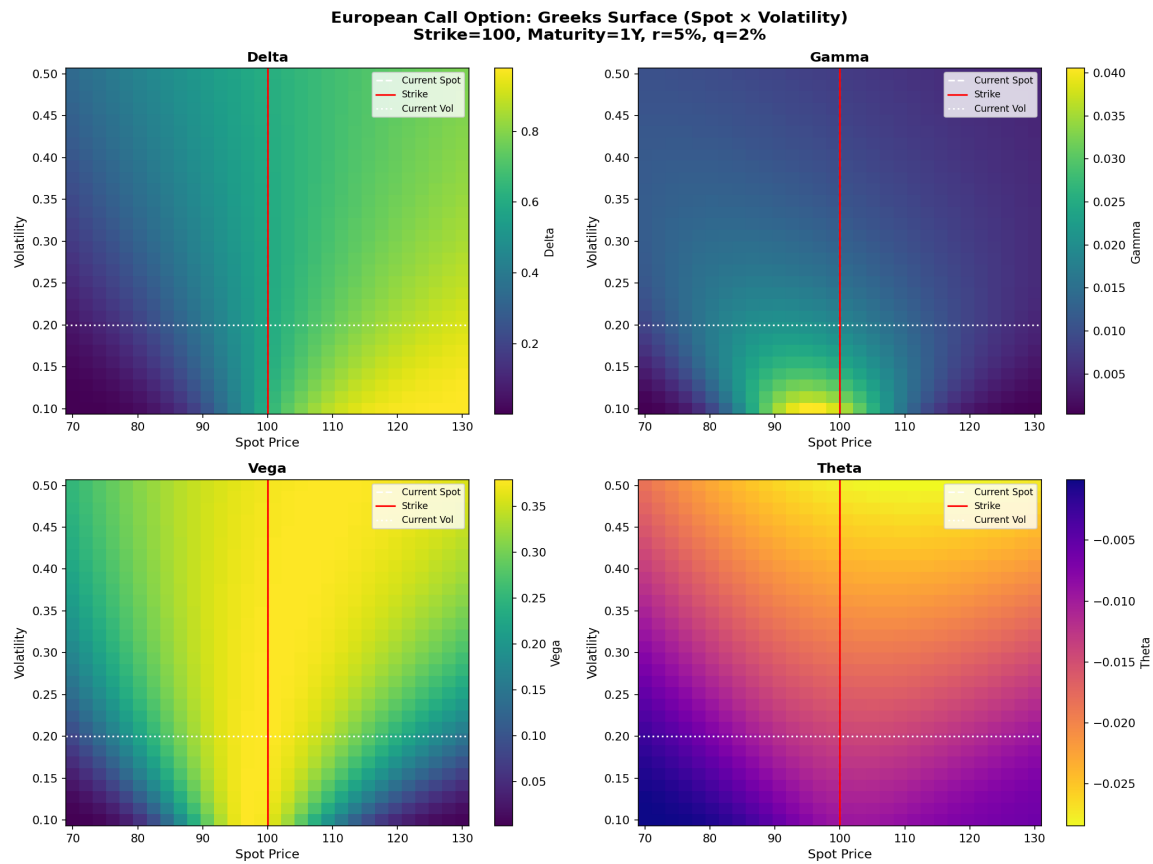
Greek	Call Option	Put Option	Interpretation
Delta	0.586851	-0.393348	Price sensitivity to \$1 change in spot
Gamma	0.018951	0.018951	Rate of change of delta (convexity)
Vega	\$0.3790	\$0.3790	Price sensitivity to 1% vol change
Theta	\$-0.0139	\$-0.0063	Time decay per day
Rho	\$0.4946	\$-0.4566	Price sensitivity to 1% rate change

5. Sensitivity Analysis

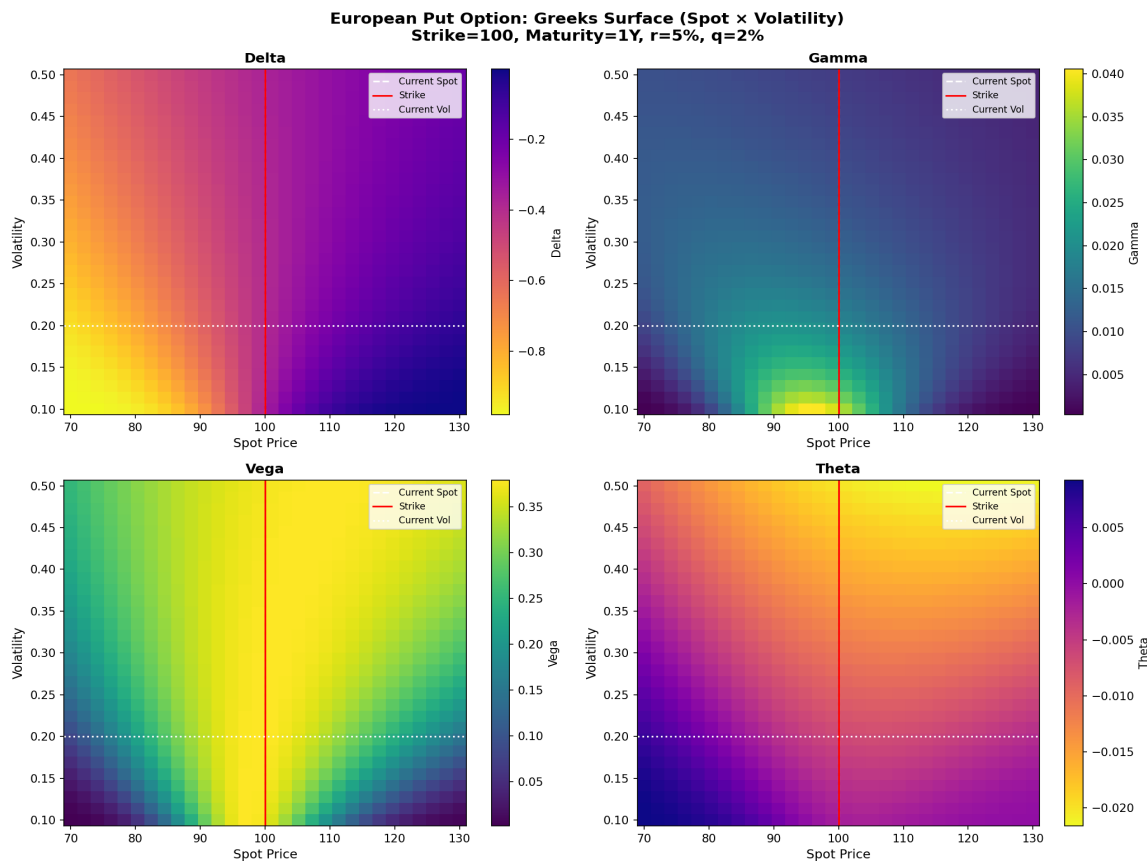
Spot × Volatility Heatmaps

The following heatmaps show how each Greek varies across different spot price and volatility levels. The red dashed line indicates the current spot price, the solid red line indicates the strike price, and the white dotted line indicates the current volatility level.

Call Option Greeks Surface



Put Option Greeks Surface



6. Key Insights

Option Valuation: Both options are **At-The-Money (ATM)** with spot equal to strike (\$100). Call premium (\$9.23) exceeds put premium (\$6.33) due to the 2% dividend yield. Time value dominates (\$6.33) for both options given the 1-year maturity. **Greek Analysis:** **Delta:** Call delta (0.587) indicates the call price will increase by approximately \$0.59 for a \$1 increase in spot. Put delta is negative (-0.393). **Gamma:** Identical for both options (0.0190), reflecting the shared convexity property. **Vega:** Both options have identical vega (\$0.38), indicating equal sensitivity to volatility changes. **Theta:** Call theta (-0.0139) is more negative than put theta (-0.0063), reflecting faster time decay for calls when dividends are present. **Rho:** Call rho is positive (\$0.49), put rho is negative (\$-0.46), as expected. **Risk Management Implications:** Delta-neutral hedge would require selling 0.587 shares per call option held (per contract). Gamma risk is highest near the strike, requiring frequent rebalancing for delta-hedged positions. Vega exposure is significant — consider volatility hedges if holding naked options.

7. Methodology

Pricing Engine: Black-Scholes-Merton Analytical Engine. **Greeks Method:** Analytical formulas (closed-form solutions). **Key Formulas:** **Call Price:** $C = S \cdot e^{(-qT)} \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2)$ **Put Price:** $P = K \cdot e^{(-rT)} \cdot N(-d2) - S \cdot e^{(-qT)} \cdot N(-d1)$ Where $d1 = [\ln(S/K) + (r - q + \sigma^2/2)T] / (\sigma\sqrt{T})$ and $d2 = d1 - \sigma\sqrt{T}$. **Model Assumptions:** Log-normal distribution of asset prices. Constant volatility and interest rates. Continuous dividend yield. European-style exercise (only at maturity). No transaction costs or taxes. Frictionless markets with infinite liquidity.